

Back Upgliding Vowels in Earlier Baltimore English

Margaret E. L. Renwick • Aidan Malanoski
Shaily Mistry • Mashal Nawabi • Yasmin Roach

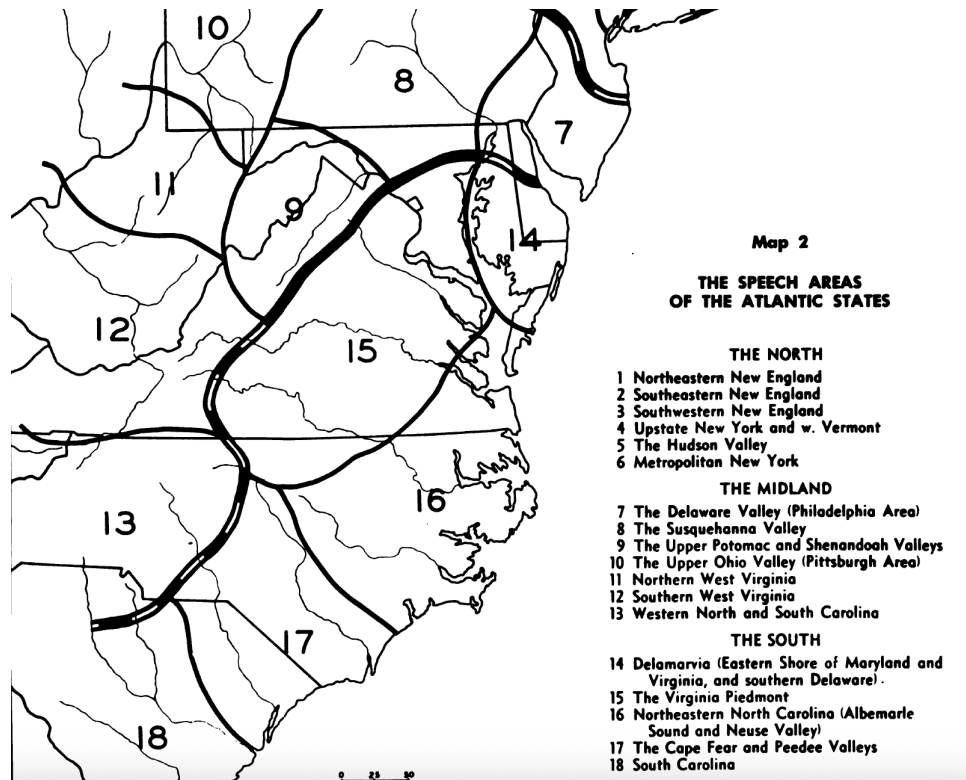
10 January 2025

Local distinctness in Charm City

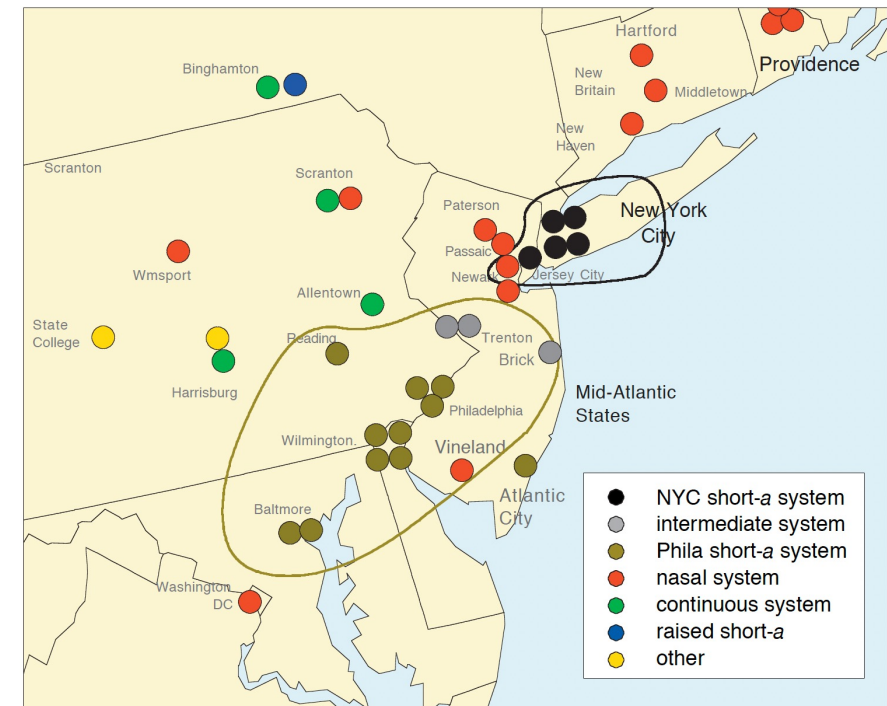
- What constellations of spoken-language features characterize speakers from Baltimore, Maryland, and how have they changed?
- “Charm City” lies near the cultural and linguistic boundary between the North and South
- We reconsider the phonetics of three back (up)gliding vowels GOOSE, GOAT, MOUTH to reveal their local distinctness among Black and white Baltimoreans.

Baltimore's regional context

Pronunciation of English in the Atlantic States (PEAS; Kurath & McDavid 1961)

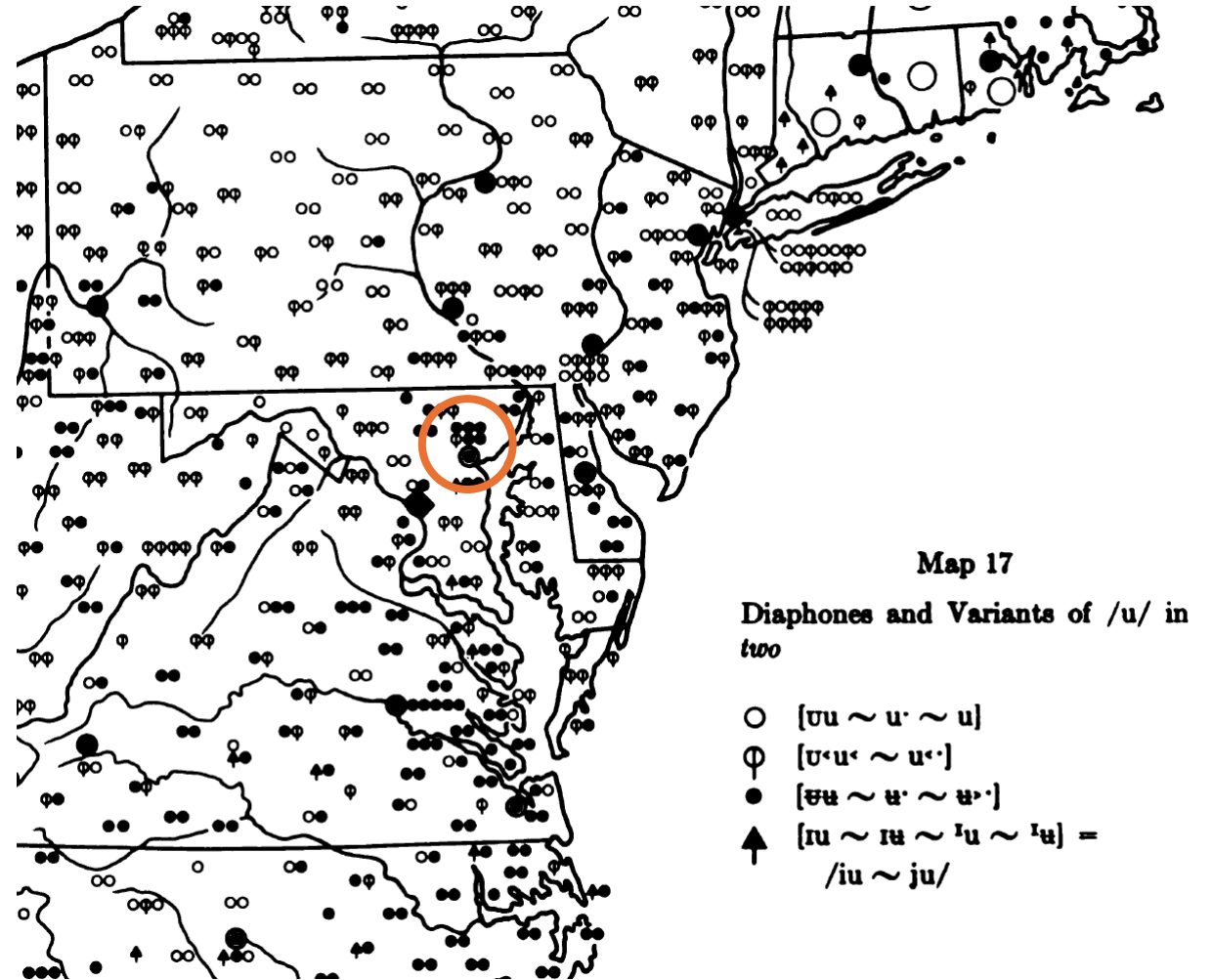


Atlas of North American English (ANAE; Labov et al. 2006)



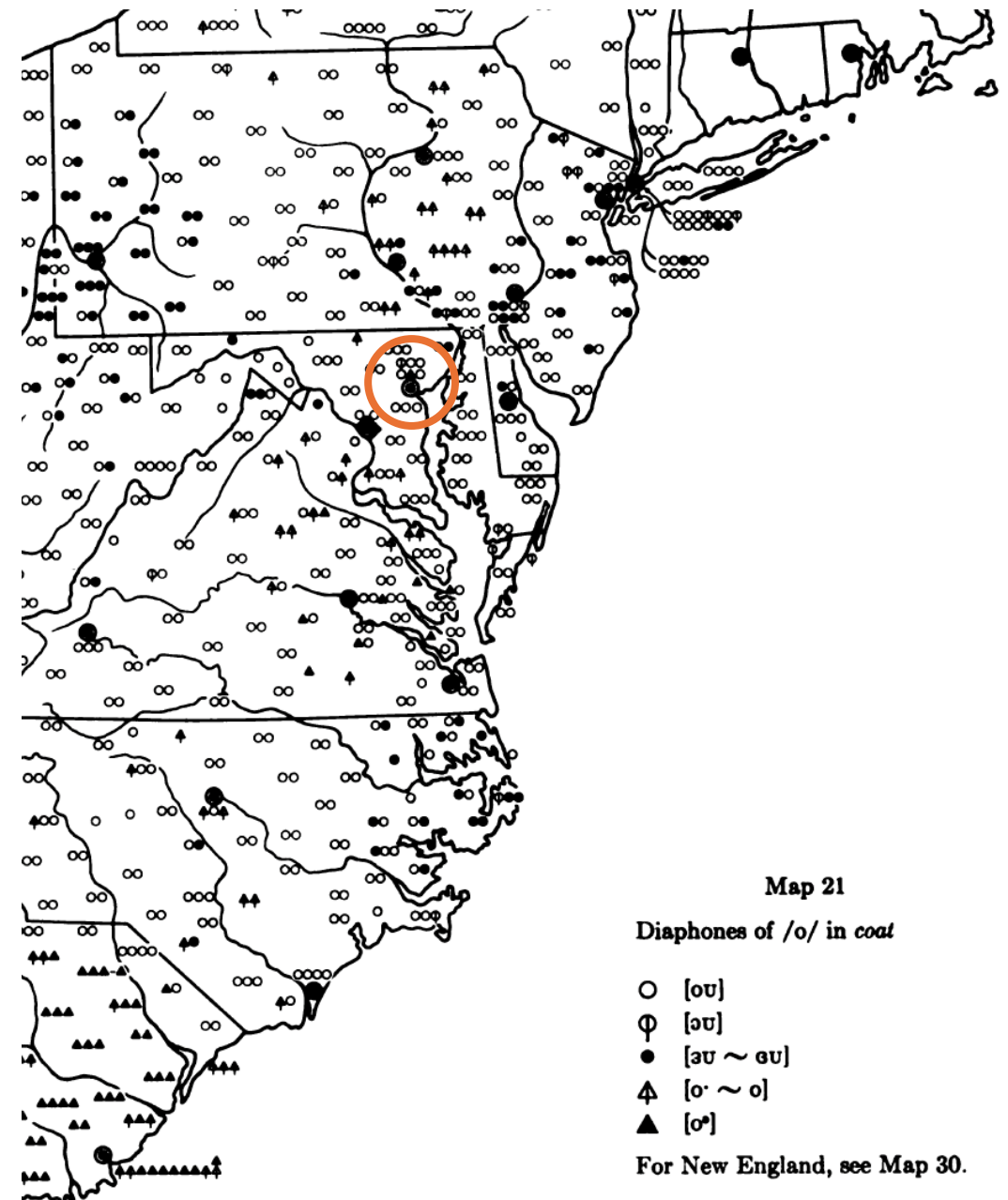
GOOSE in Baltimore

- In PEAS:
 - central
- In ANAE:
 - front
- GOOSE-fronting is a well-known stereotype of Baltimore's AAE



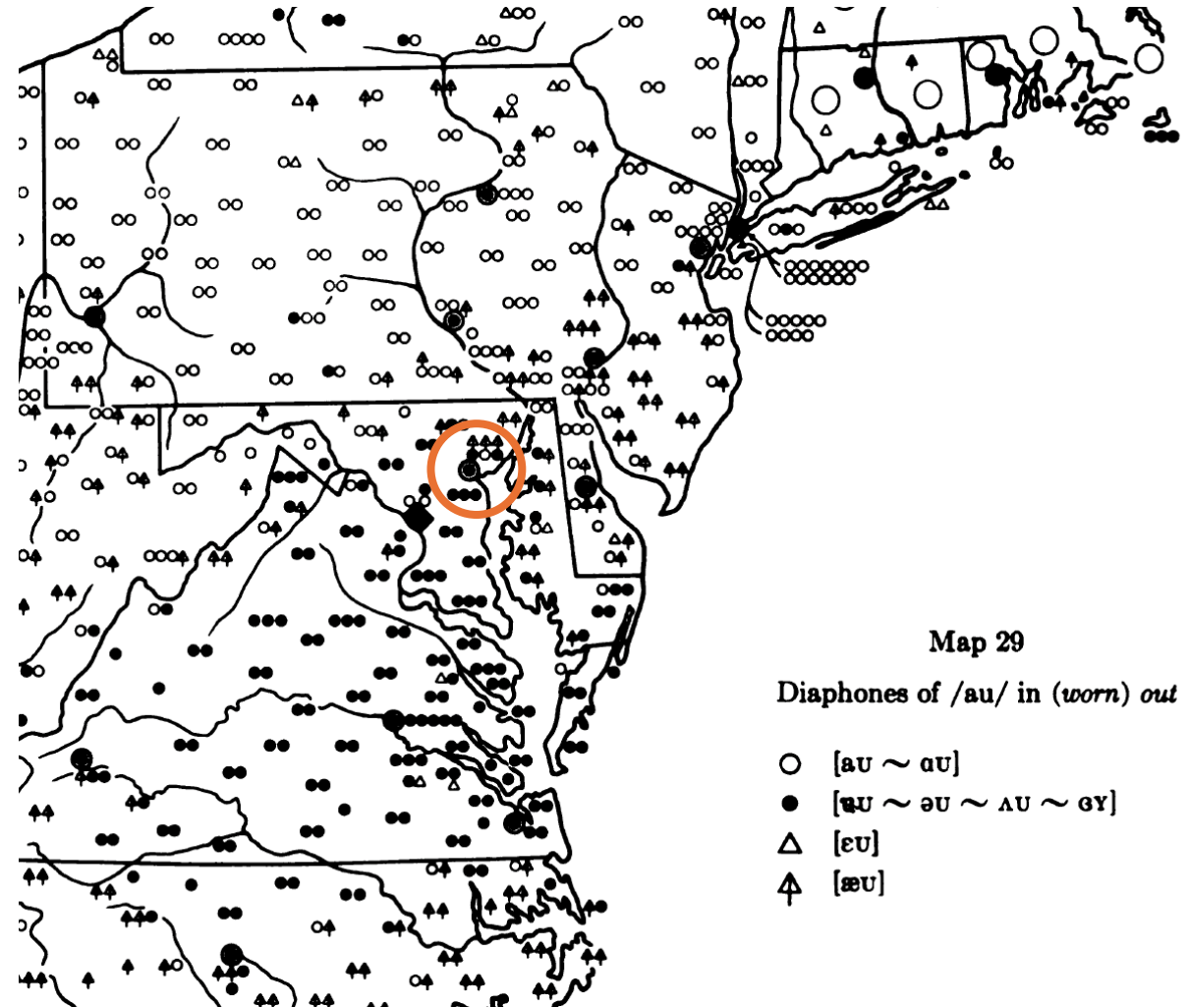
GOAT in Baltimore

- In PEAS:
 - back
 - fronted in “cultivated” speech
- In ANAE:
 - fronted



MOUTH in Baltimore

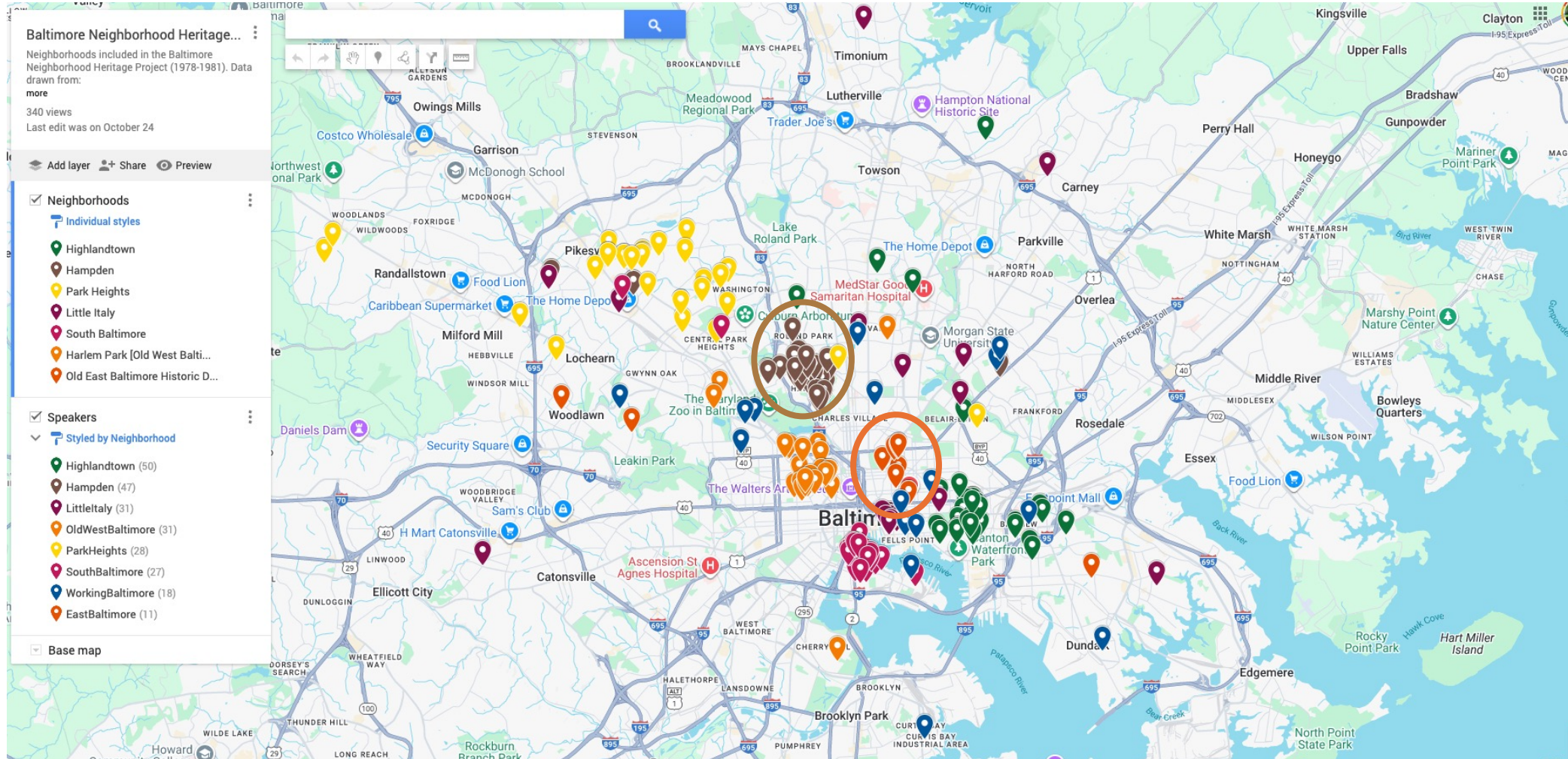
- In PEAS:
 - fronted
 - “Canadian” raising occurs sometimes before voiceless consonants
- In ANAE:
 - fronted and raised along front diagonal



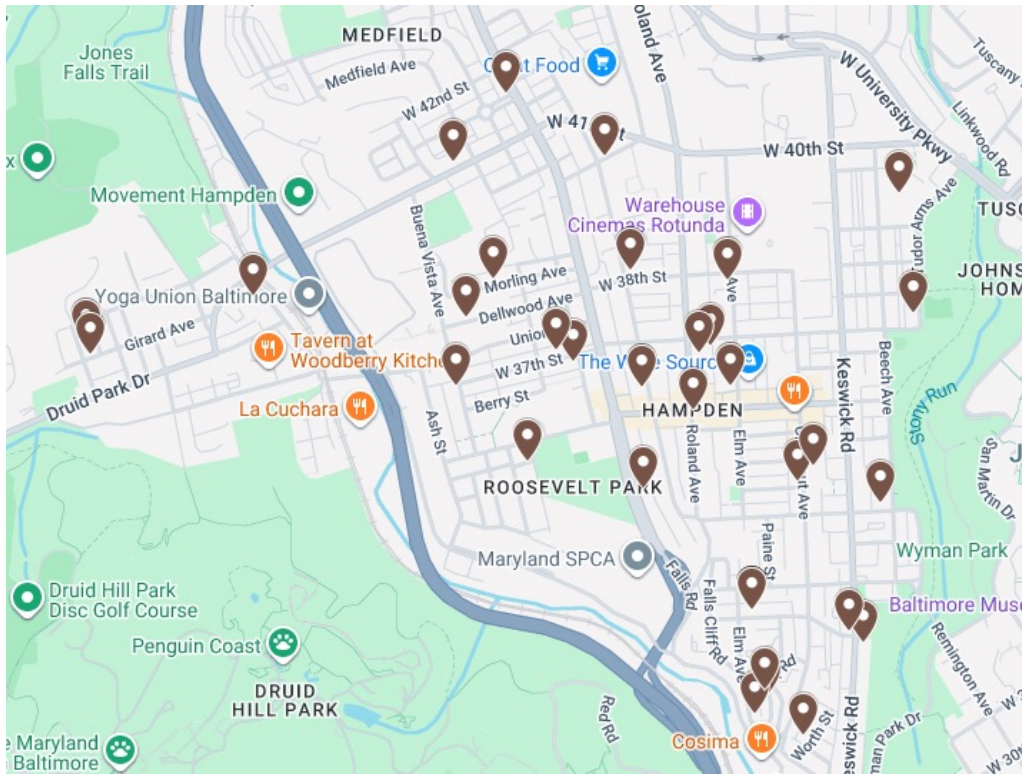
Baltimore Neighborhood Heritage Project

- Community-based oral histories gathered 1978–1981
 - Goal: to recover and present the history of seven working-class and ethnic neighborhoods in Baltimore City, thus democratizing the historical record (Shopes 1981)
- 232 speakers, born 1879–1951 (median y.o.b. 1909)
 - Many interviews transcribed, beginning in the 1980s
 - Audio and text digitized by the University of Baltimore’s Special Collections & Archives
- Neighborhoods and collections:
 - **East Baltimore** (Black, ~Protestant)
 - Old West Baltimore (Black, ~Protestant)
 - **Hampden** (white, ~Protestant)
 - Highlandtown (mostly white, ~Catholic)
 - Little Italy (white, ~Catholic)
 - Park Heights (then mostly white, mostly Jewish)
 - South Baltimore (Black and white, various religions)
 - “Working Baltimore” (Black and white men, various religions)

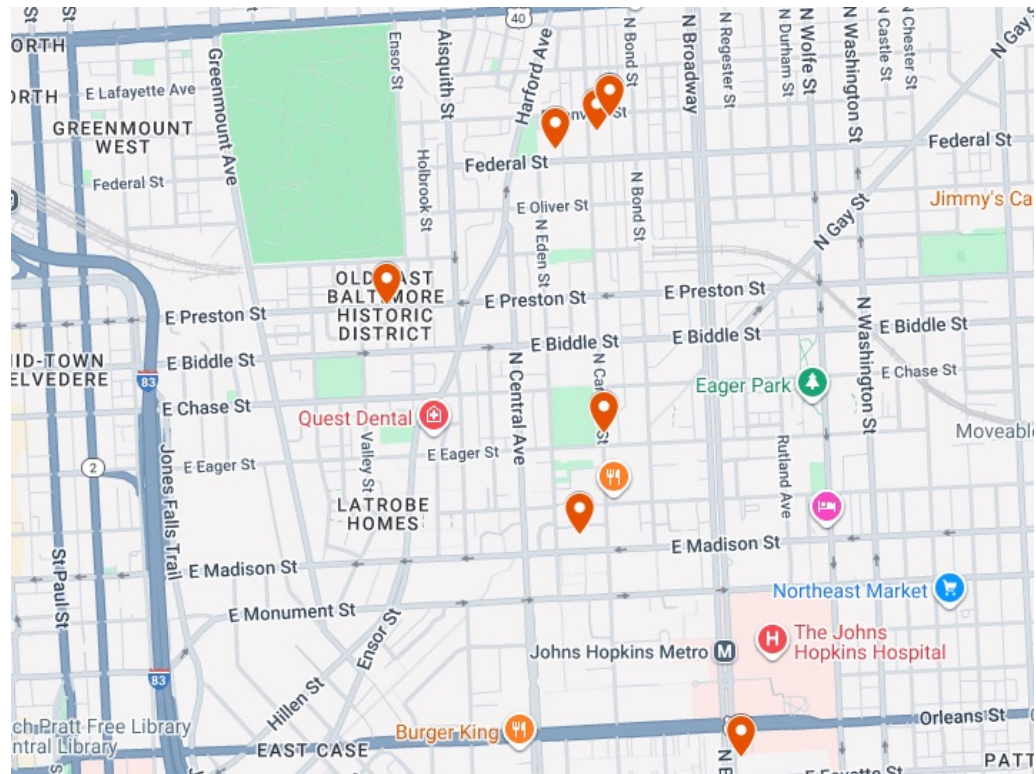
Mapping the BNHP's participants



Hampden (+ Woodberry)



East Baltimore



Interviewees analyzed here

East Baltimore (Black)

Name	gender	yob	age	education
Mary Tunstall	F	1893	85	11-12 years
Bessie Queen	F	1900	78	0-7 years
Mattie Spencer	F	1900	78	13-16+ years
Margaret Proctor	F	1904	74	0-7 years
Inez Royster	F	1905	73	11-12 years
Myrtle Smith	F	1907	71	11-12 years
Odessa Thomas	F	1911	67	11-12 years
Anita Horne	F	1918	60	11-12 years
Harry Queen	M	1901	77	0-7 years

Hampden (White)

Name	gender	yob	age	education
Elmira Deitrich	F	1896	83	13-16+ years
Mary Proctor	F	1902	77	0-7 years
Hester Worden	F	1903	76	0-7 years
Beatrice Reigle	F	1907	72	11-12 years
Helen Lindsey	F	1914	65	11-12 years
Winfield Lytle	M	1908	71	8-10 years
Richard Meads	M	1910	69	0-7 years
Albert Arnold	M	1917	62	8-10 years
Carl Burke	M	1935	44	13-16+ years

Data analysis pipeline for BNHP

- BNHP files downloaded from UBalt Archive: WAV & PDF
 - Audio converted to mono → diarization with Bleaman & Sprouse's toolkit (2023) to produce .eaf files indicating which speaker is talking
 - Transcriptions extracted from PDFs with `tesseract` → .txt form (messy!)
- Unification in ELAN
 - Diarization output opened in ELAN, and text of transcription is inserted at the utterance level; utterance boundaries adjusted as needed
 - Files are converted to TextGrid and checked for accuracy after unification
- Forced alignment with MFA
- Formant extraction with DARLA
 - Provides nucleus values and formants at 20%, 35%, 50%, 65%, 80%

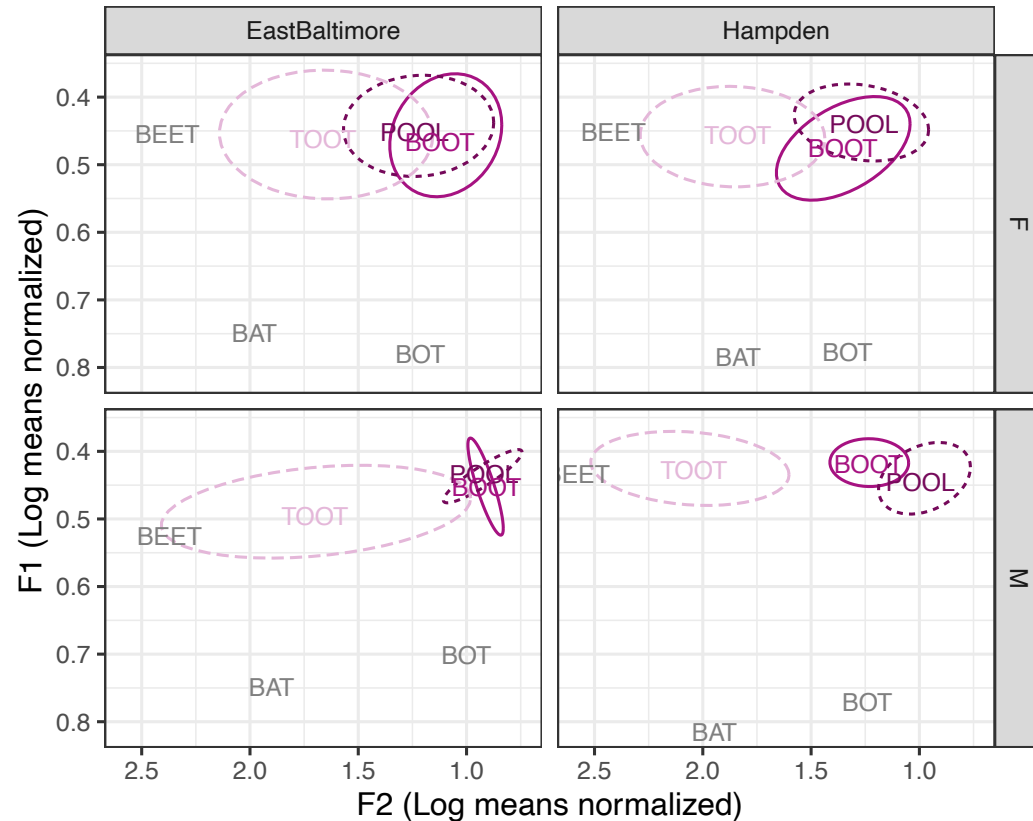
Data processing and token counts

- Data processed according to Stanley's (2023) order of operations
 - Tokens lacking incomplete trajectory data are removed
 - Outliers are removed using Mahalanobis distance
 - Normalization (i.e. Log Means)
 - Stop words are removed

Lexical set	East Baltimore	Hampden
GOOSE	316	668
GOAT	816	1688
MOUTH	244	518
Total tokens	1376	2874

GOOSE and its allophones

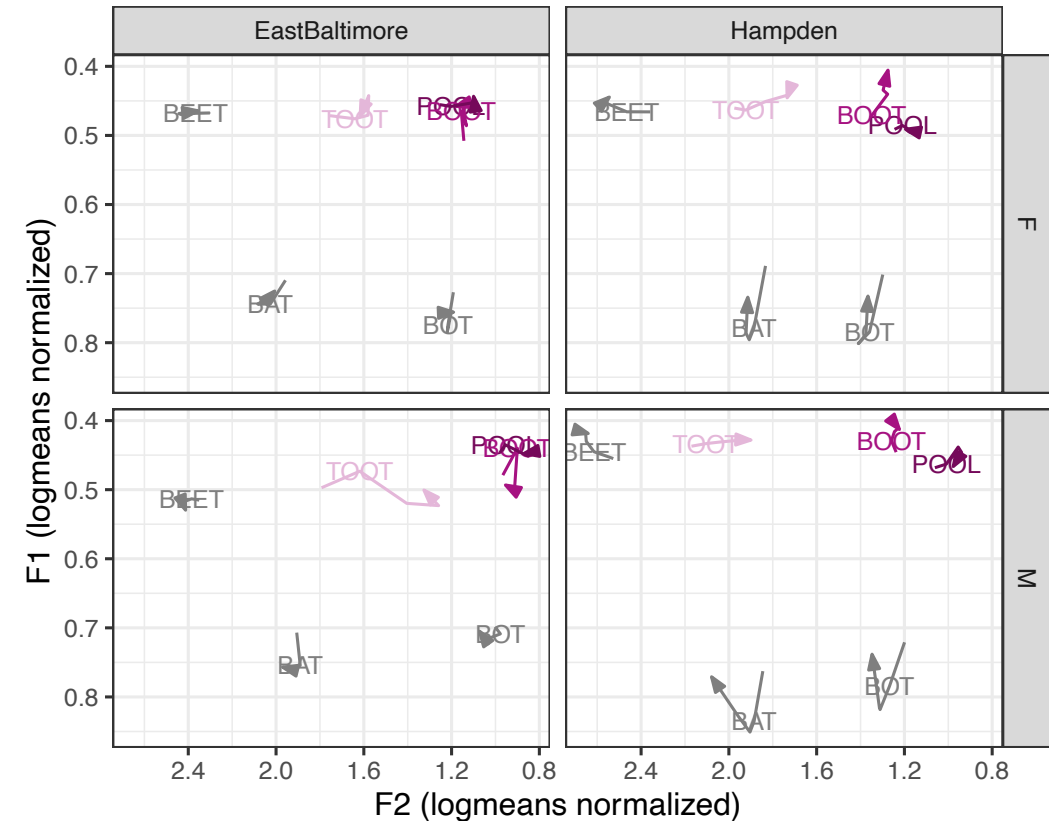
Nuclei of GOOSE by neighborhood



Allophone

a BAT	α BOOT	α POOL
a BEET	a BOT	α TOOT

Median GOOSE trajectories by neighborhood

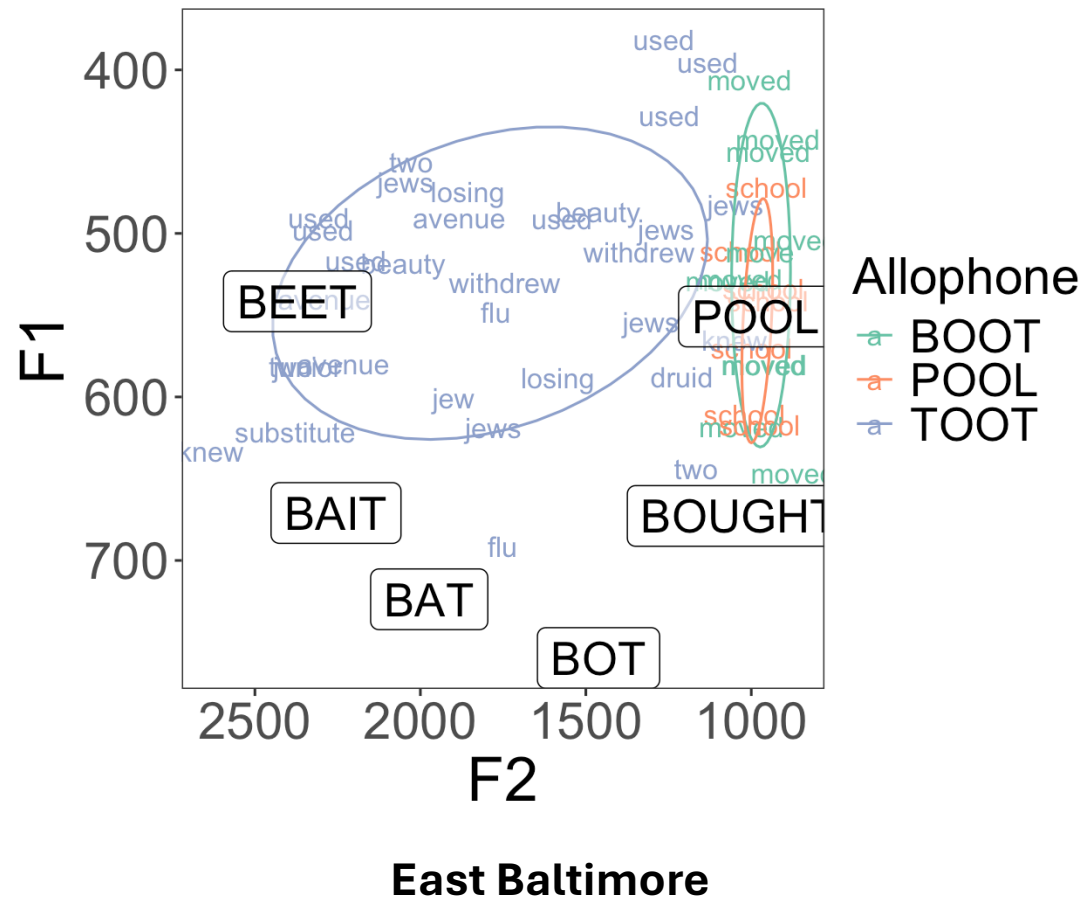


Allophone

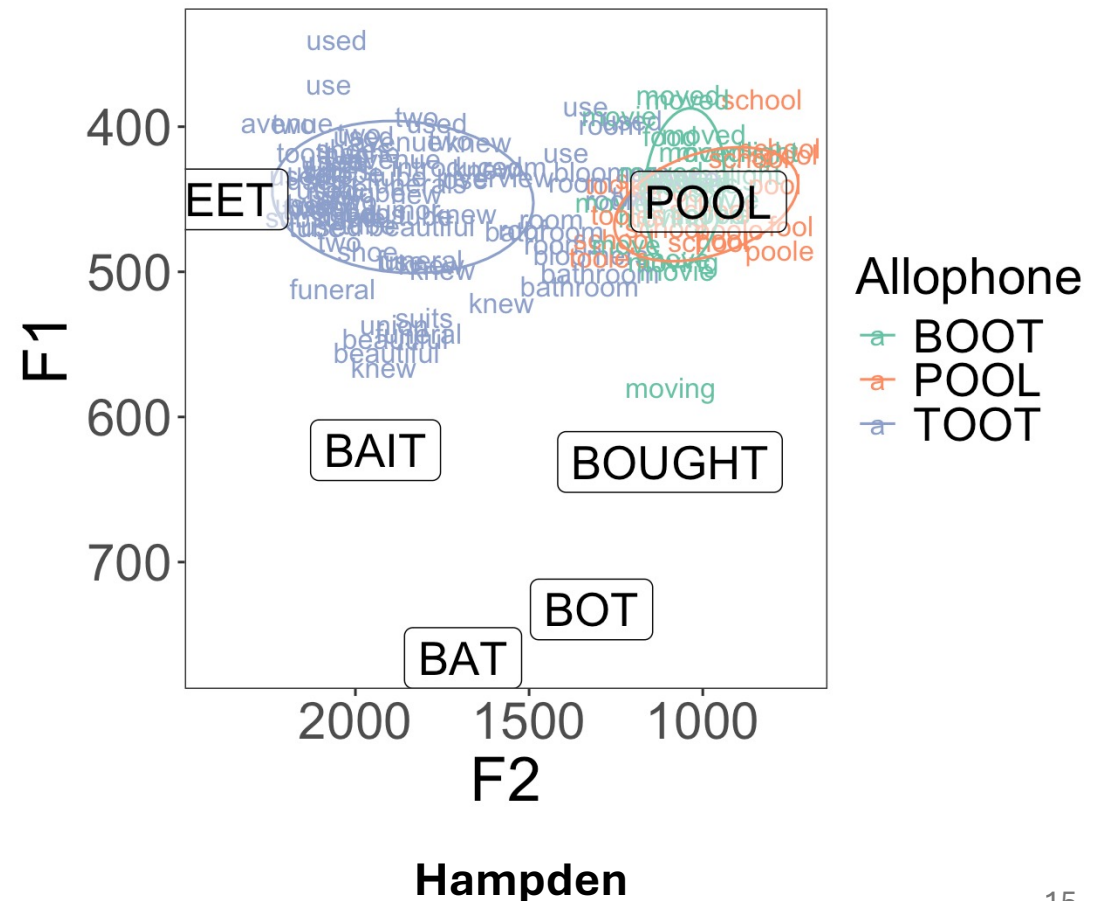
α BAT	α BOOT	α POOL
α BEET	α BOT	α TOOT

GOOSE: Representative speakers

Myrtle Smith: GOOSE



Hester Worden: GOOSE



GOOSE: Audio samples



Myrtle Smith: Well, for two months, I tried housework.



Myrtle Smith: ...from there, we went up to um– high school on Dolphin and um– Pennsylvania Avenue.



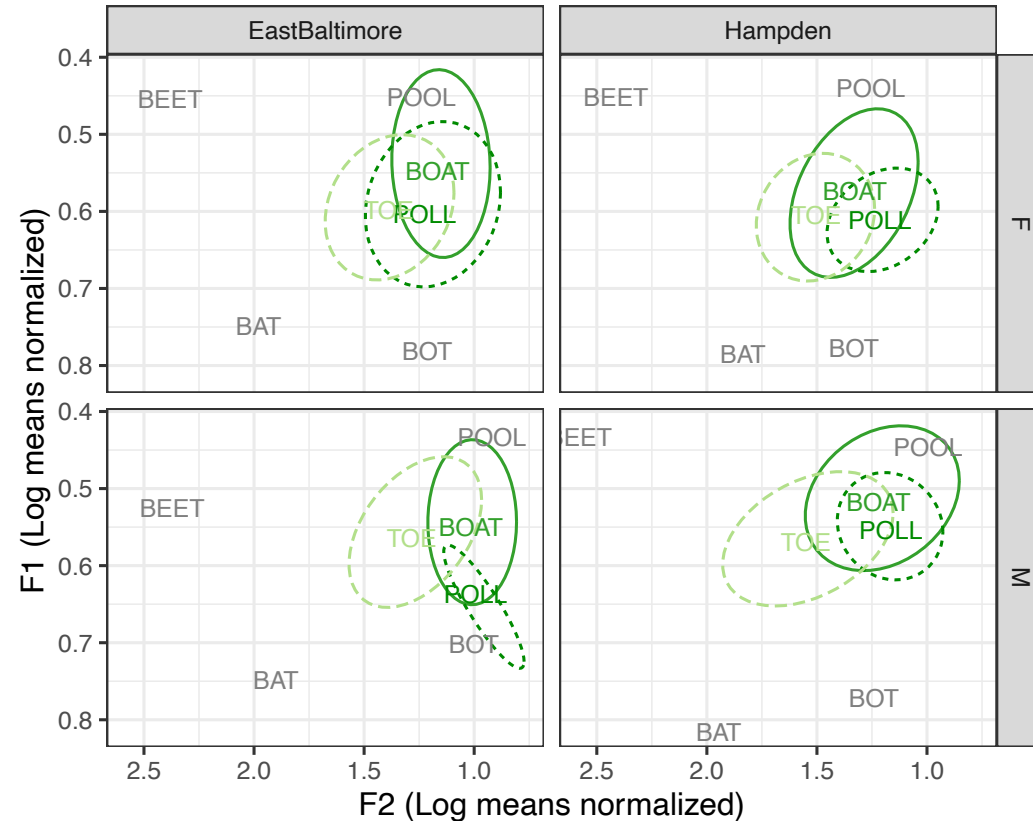
Hester Worden: So for two days I couldn't move it.



Hester Worden: I'd come home from school or sit on the avenue.

GOAT and its allophones

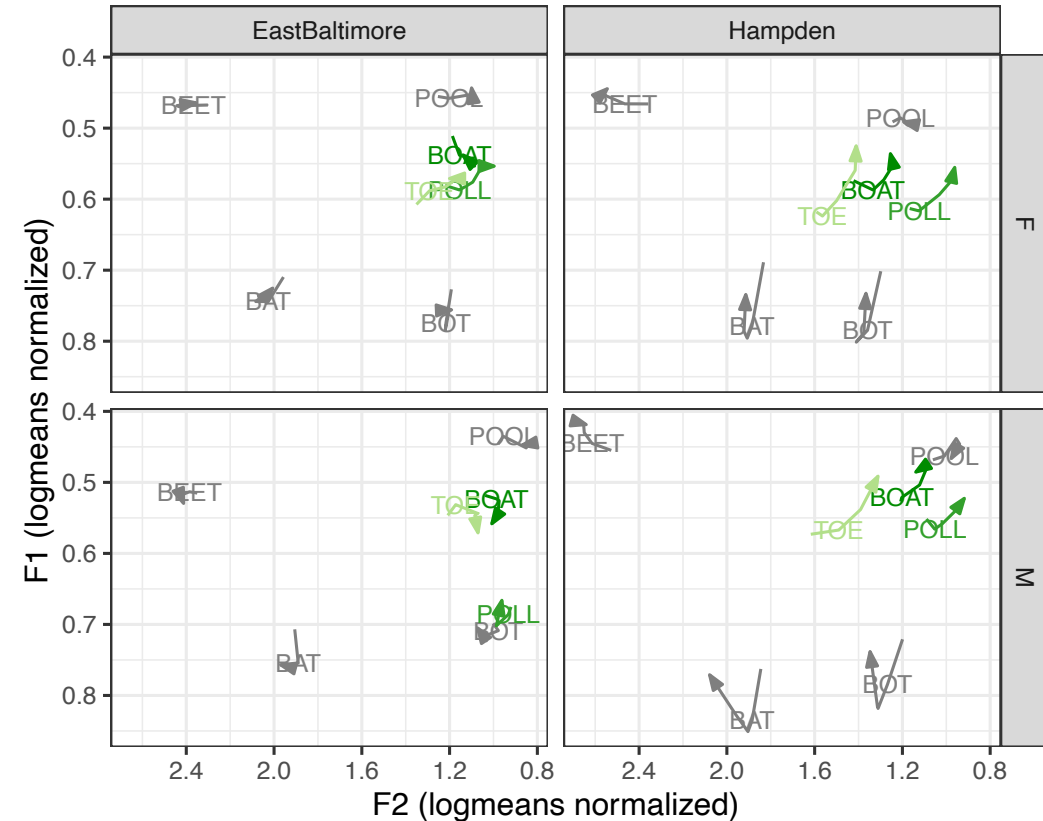
Nuclei of GOAT by neighborhood



Allophone

a	BAT	↗	BOAT	↗	POLL	↗	TOE
a	BEET	a	BOT	a	POOL		

Median GOAT trajectories by neighborhood

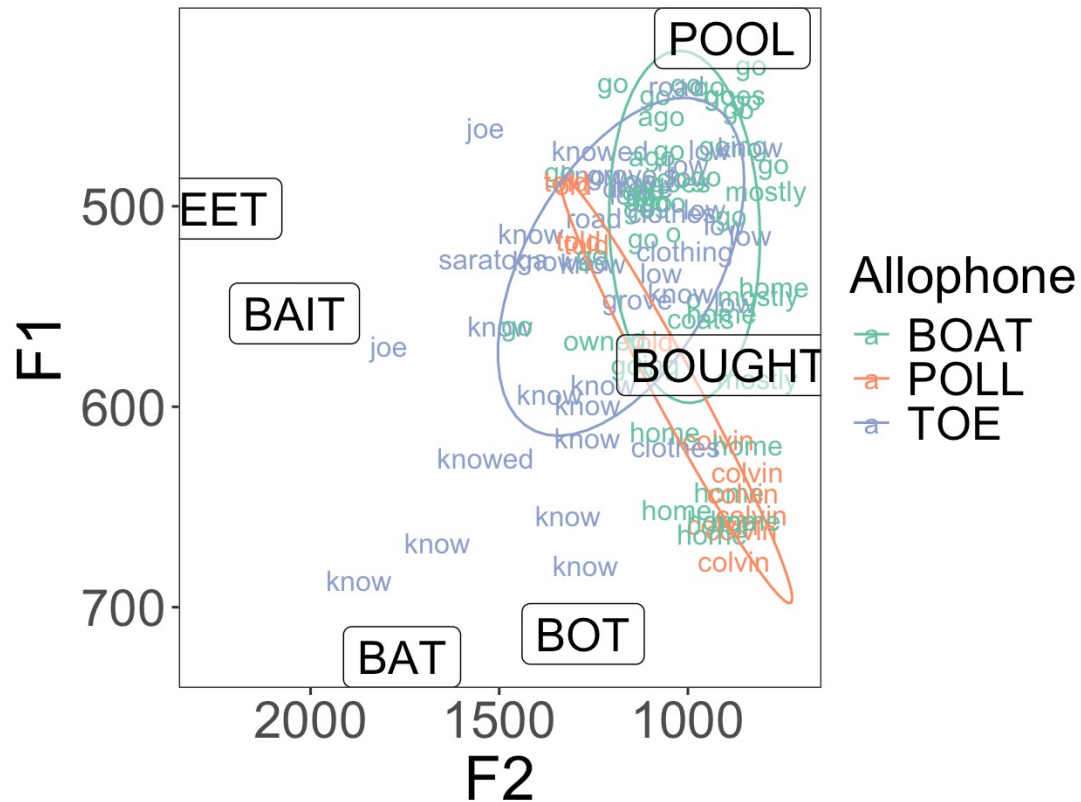


Allophone

↗	BAT	↗	BOAT	↗	POLL	↗	TOE
↗	BEET	↗	BOT	↗	POOL		

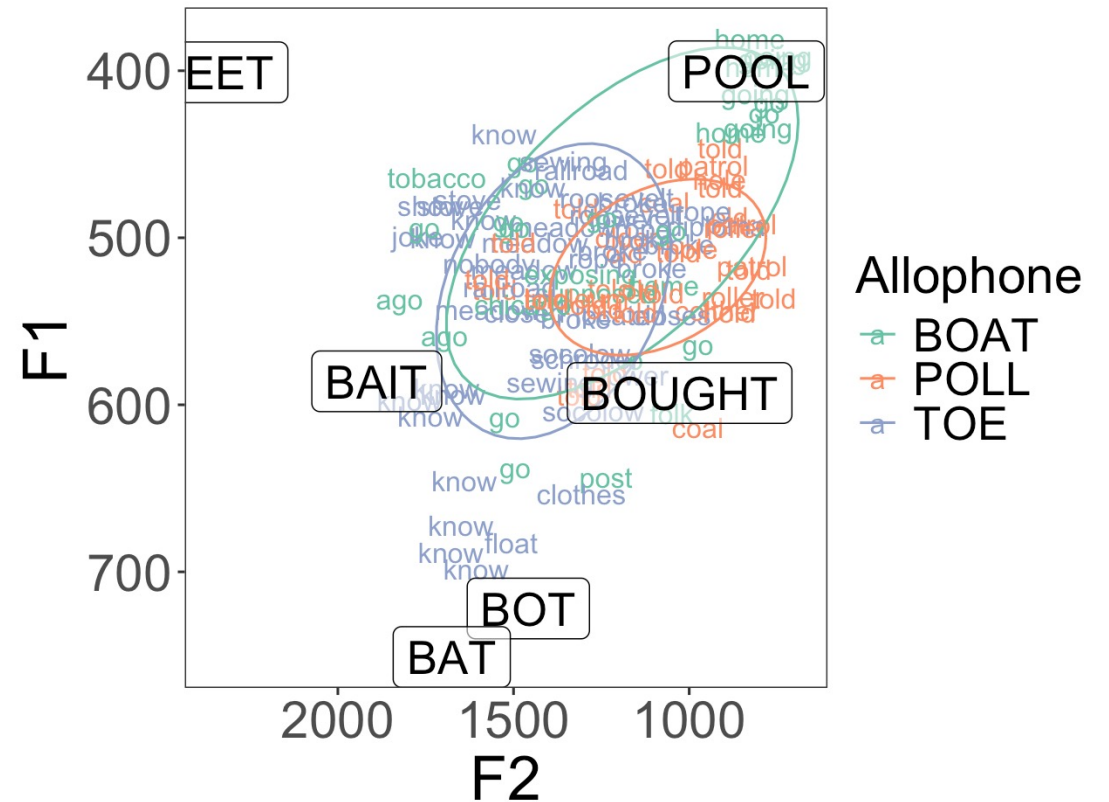
GOAT: Representative speakers

Harry Queen: GOAT



East Baltimore

Winfield Lytle: GOAT



Hampden

GOAT: Audio samples



Harry Queen: On Belair Road, yeah.



Harry Queen: See, all them polices walked the beat. They knowed everybody in that neighborhood.



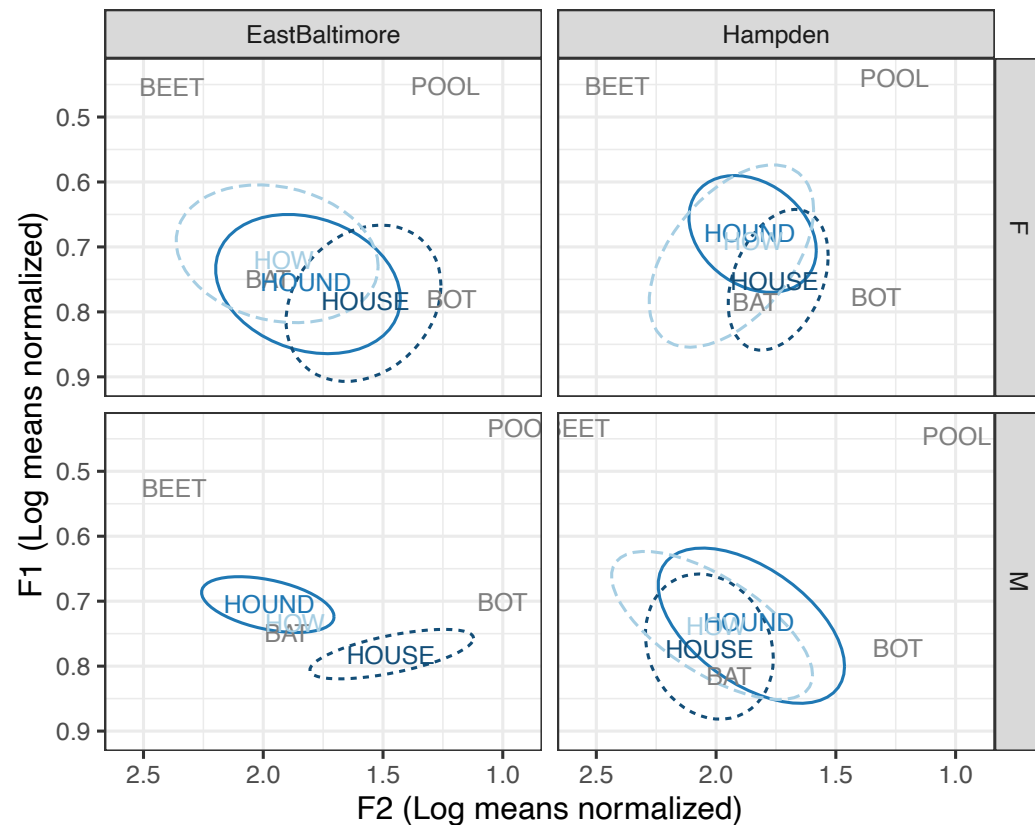
Winfield Lytle: Course they called ‘em, “bulls.” That was the police from the railroad.



Winfield Lytle: Well, you know where Tudor Arms is?

MOUTH and its allophones

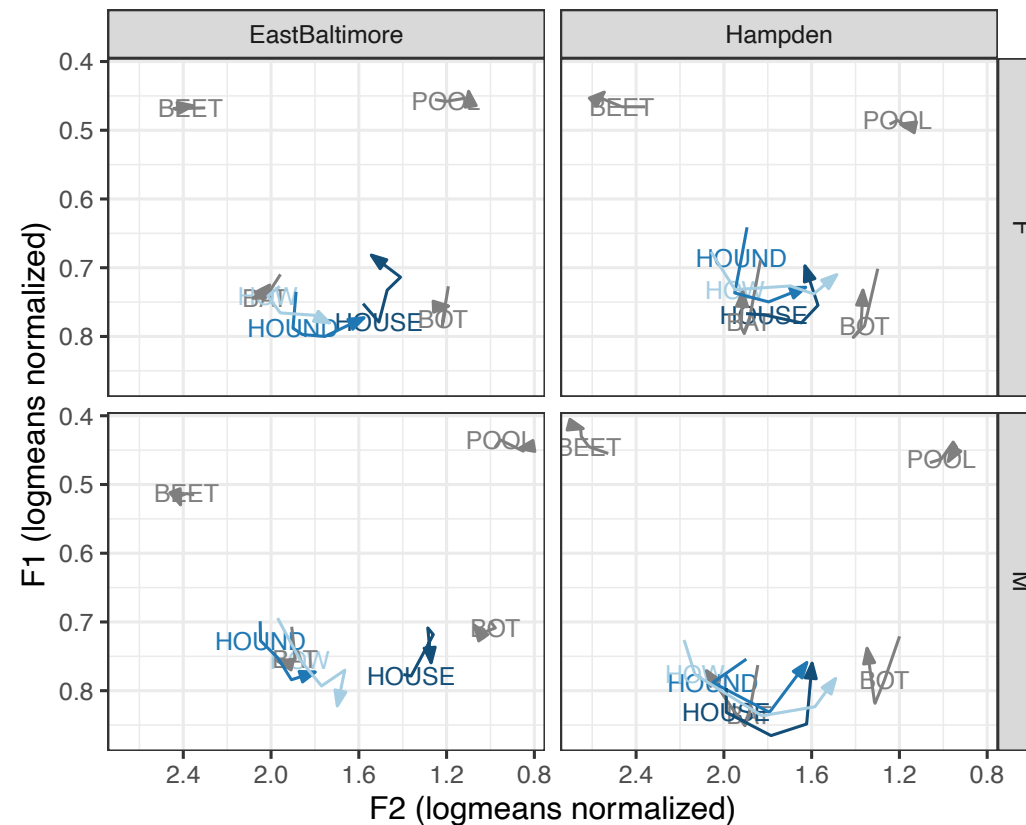
Nuclei of MOUTH by neighborhood



Allophone

a BAT	a BOT	HOUSE	a POOL
a BEET	HOUND	HOW	

Median MOUTH trajectories by neighborhood



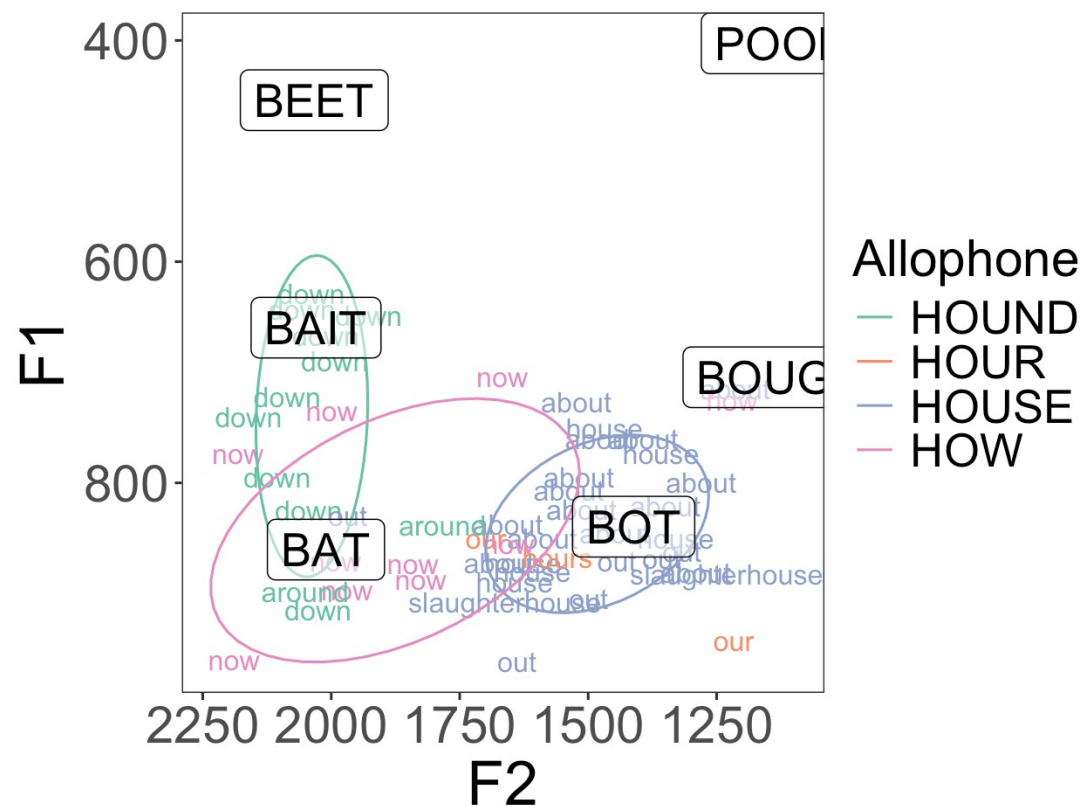
Allophone

BAT	BOT	HOUSE	POOL
BEET	HOUND	HOW	

MOUTH: Representative speakers

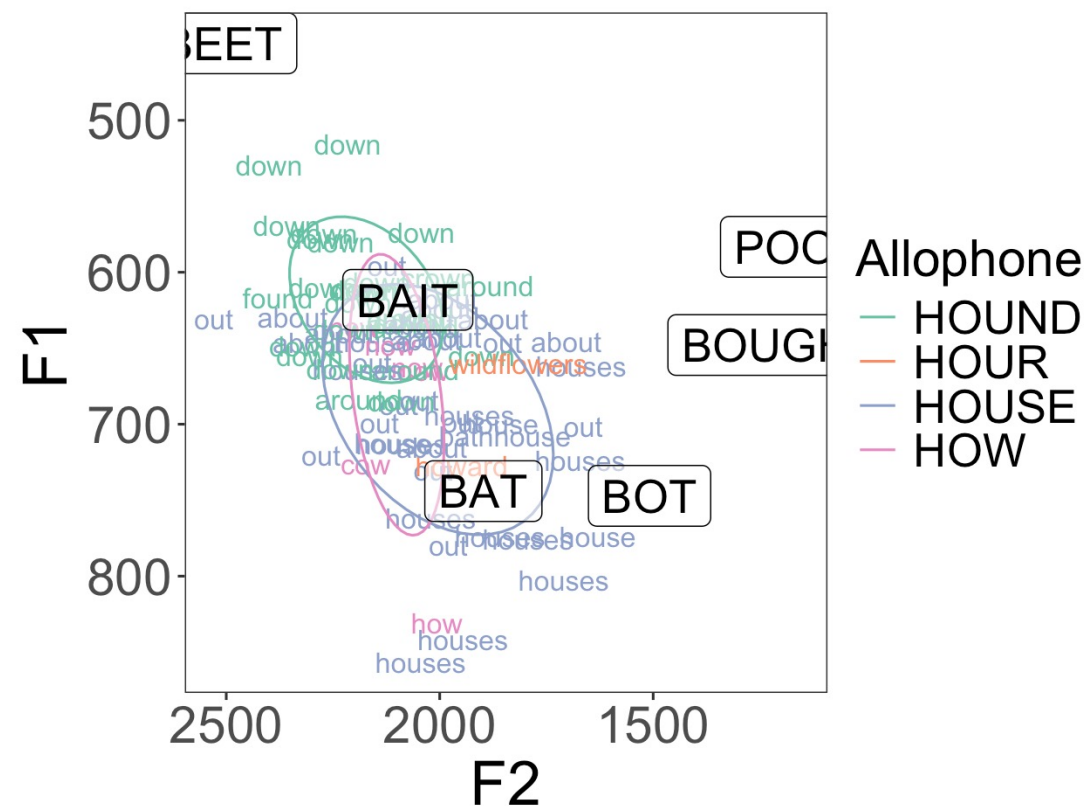
stop words included for illustration

Odessa Thomas: MOUTH



East Baltimore

Beatrice Reigle: MOUTH



Hampden

MOUTH: Audio samples



Odessa Thomas: Urban renewal came through and tore my house down.



Odessa Thomas: About... 1950-something.



Odessa Thomas: Oh, they're almost like food stamps are now.

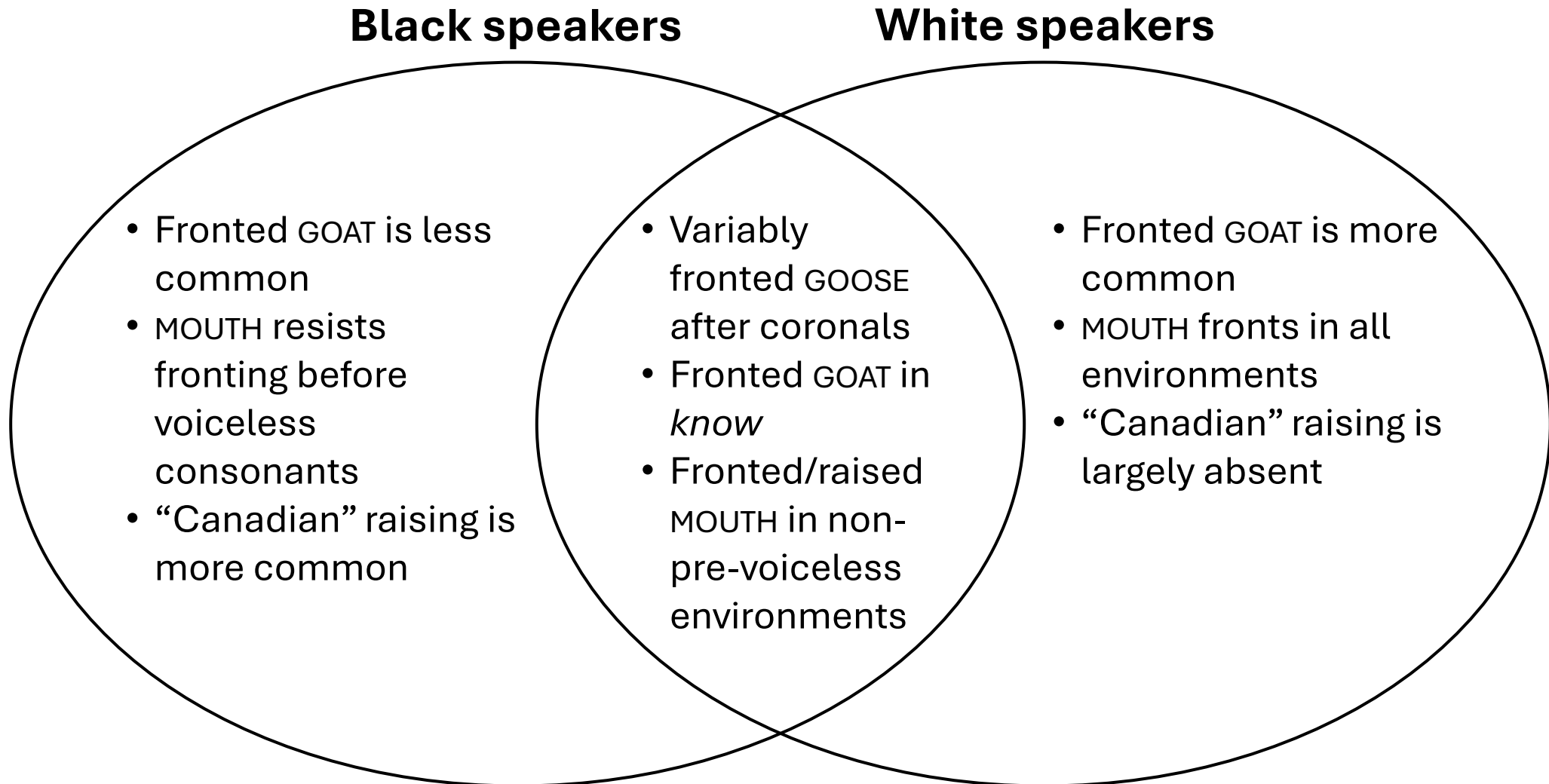


Beatrice Reigle: Oh, well, pop, he left the house to mom. He said, "Mom, r– run the house." He br– he brought the money. He made the money and brought it home and give it to her, she run the house.



Beatrice Reigle: We– talked about that for– months before the time came, and I think we had as much fun talking about it as we did going down the bay.

Summary



Discussion

- Demographic patterns may inform linguistic patterns
 - Migration
 - Seclusion and segregation
- The Black speakers' patterns echo those reported in DC (Arnson & Farrington 2017)
- The white speakers' patterns are transitional between those in PEAS and ANAE
- Future research will:
 - include speakers from more neighborhoods
 - analyze more vowels (e.g., THOUGHT, short-a)

Thank you! (“Thanks, hon!”)

- Aidan Faust at UBalt
- Linda Shopes and the BNHP team
- Shaily, Mashal, and Yasmin at JHU



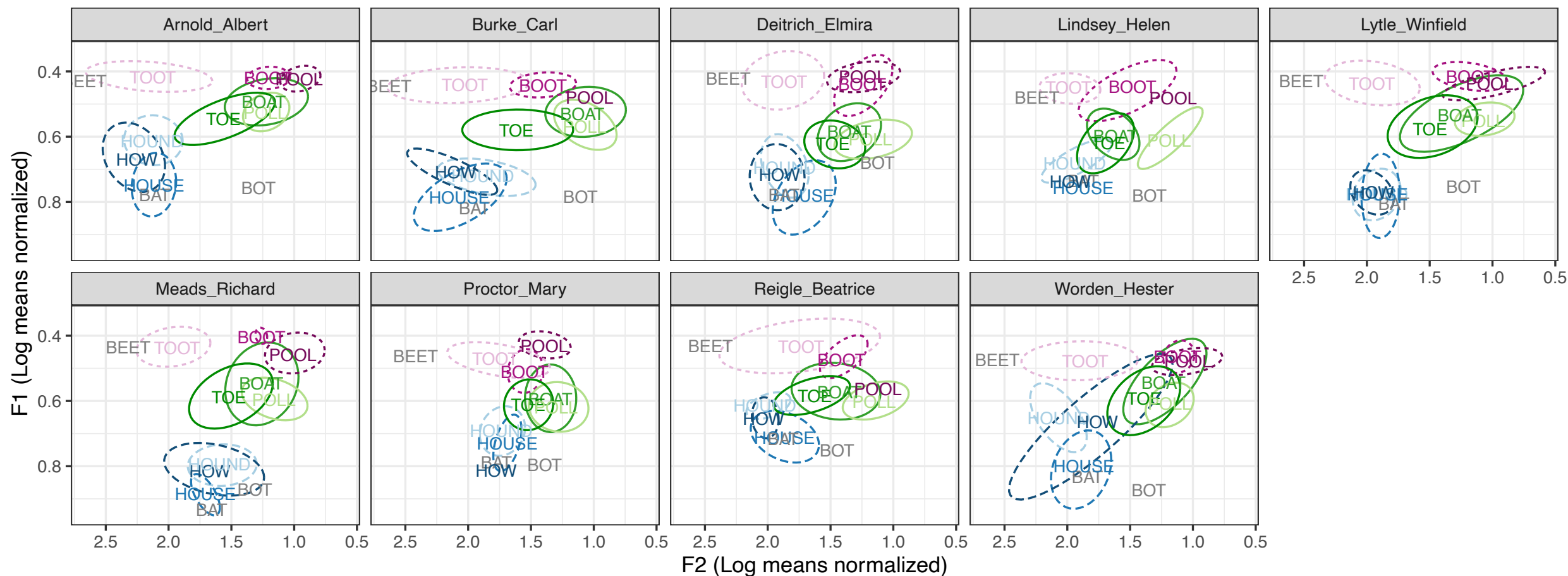
References

- Arnson, Shelby & Charlie Farrington. 2017. Twentieth Century Sound Change in Washington DC African American English. *University of Pennsylvania Working Papers in Linguistics* 23(2). 1–10.
- Baltimore Neighborhood Heritage Project. University of Baltimore Special Collections and Archives. <https://archivesspace.ubalt.edu/repositories/2/resources/38>. (8 July, 2025).
- Bleaman, Isaac L. & Ronald L. Sprouse. 2023. Speaker Diarization for Linguistics. *Linguistics Methods Hub*. <https://doi.org/10.5281/zenodo.7747131>.
- Kenyon, John S. & Thomas A. Knott. 1949. *A Pronouncing Dictionary of American English*. Springfield, MA: G. & C. Merriam Company.
- Kurath, Hans & Raven I. Jr. McDavid. 1961. *The Pronunciation of English in the Atlantic States: Based upon the Collections of the Linguistic Atlas of the Eastern United States*. Ann Arbor, MI: University of Michigan Press.
- Labov, William, Sharon Ash & Charles Boberg. 2006. *The Atlas of North American English: Phonetics, phonology, and sound change: A multimedia reference tool*. Berlin; New York: Mouton de Gruyter.
- Labov, William & Sabriya Fisher. 2015. African American phonology in a Philadelphia community. In Sonja L. Lanehart (ed.), *The Oxford handbook of African American language*, 256–279. New York: Oxford University Press.
- Labov, William, Ingrid Rosenfelder & Josef Fruehwald. 2013. One hundred years of sound change in Philadelphia: Linear incrementation, reversal, and reanalysis. *Language* 89(1). 30–65. <https://doi.org/10.1353/lan.2013.0015>.
- Lee, Sinae. 2016. High and Mid Back Vowel Fronting in Washington, D.C. *American Speech* 91(4). 425–471. <https://doi.org/10.1215/00031283-3870152>.
- Malanoski, Aidan. 2024. *A report on /oy/-monophthongization in Baltimore*. Paper presented at the Annual Meeting of the American Dialect Society, New York, NY.
- McAuliffe, Michael, Michaela Socolof, Sarah Mihuc, Michael Wagner & Morgan Sonderegger. 2017. Montreal Forced Aligner: trainable text-speech alignment using Kaldi. In *Proceedings of Interspeech 2017*, 498–502. Stockholm, Sweden. https://www.isca-archive.org/interspeech_2017/mcauliffe17_interspeech.pdf.
- Pietila, Antero. 2010. *Not in my neighborhood*. Chicago: Ivan R. Dee.
- Reddy, Sravana & James N. Stanford. 2015. Toward completely automated vowel extraction: Introducing DARLA. *Linguistics Vanguard* 1(1). 15–28. <https://doi.org/10.1515/lingvan-2015-0002>.
- Shopes, Linda. 1981. The Baltimore Neighborhood Heritage Project: Oral History and Community Involvement. *Radical History Review* 1981(25). 27–44. <https://doi.org/10.1215/01636545-1981-25-27>.
- Shopes, Linda. 1986. Oral history and community involvement: the Baltimore Neighborhood Heritage Project. In Susan Porter Benson, Steven Brier & Roy Rosenzweig (eds.), *Presenting the past: essays on history and the public* (Critical Perspectives on the Past), 249–263. Philadelphia: Temple University Press.
- Stanley, Joseph A. 2022. Interpreting the order of operations in a sociophonetic analysis. *Linguistics Vanguard*. De Gruyter Mouton 8(1). 279–289. <https://doi.org/10.1515/lingvan-2022-0065>.

Hampden speakers: nuclei

Vowel nuclei by speaker and by allophone

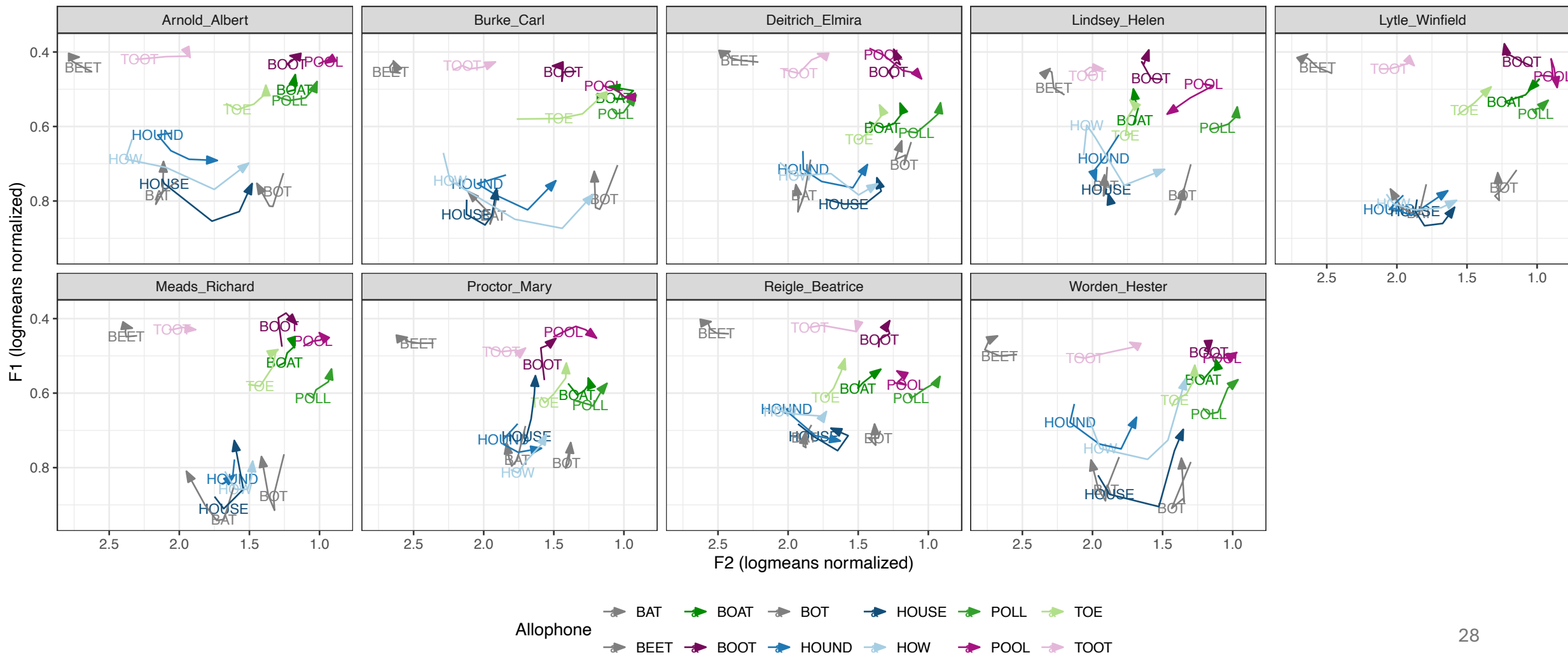
Hampden speakers



Hampden speakers: trajectories

Median trajectories by speaker and by allophone

Hampden speakers



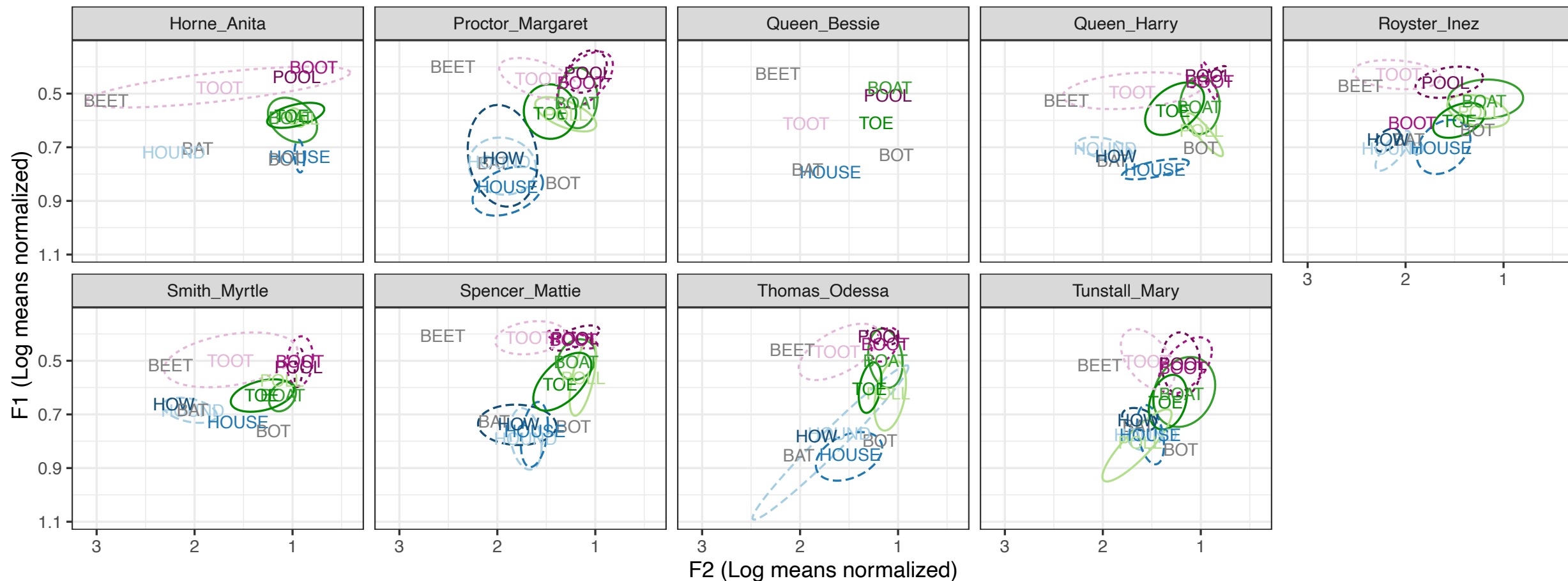
Hampden: more details (interviews 1978)

Name	gender	yob	age	education	ethnicity	birthplace	religion
Elmira Deitrich	F	1896	83	13-16+ years	White	Baltimore	Methodist
Mary Proctor	F	1902	77	0-7 years	White	VA	Christian
Hester Worden	F	1903	76	0-7 years	White	Baltimore	Methodist
Beatrice Reigle	F	1907	72	11-12 years	White	Baltimore	Methodist
Helen Lindsey	F	1914	65	11-12 years	White	Baltimore	Methodist
Winfield Lytle	M	1908	71	8-10 years	White	Baltimore	Methodist
Richard Meads	M	1910	69	0-7 years	White	MD	Episcopal
Albert Arnold	M	1917	62	8-10 years	White	Baltimore	Methodist
Carl Burke	M	1935	44	13-16+ years	White	Baltimore	Presbyterian

East Baltimore speakers: nuclei

Vowel nuclei by speaker and by allophone

East Baltimore speakers

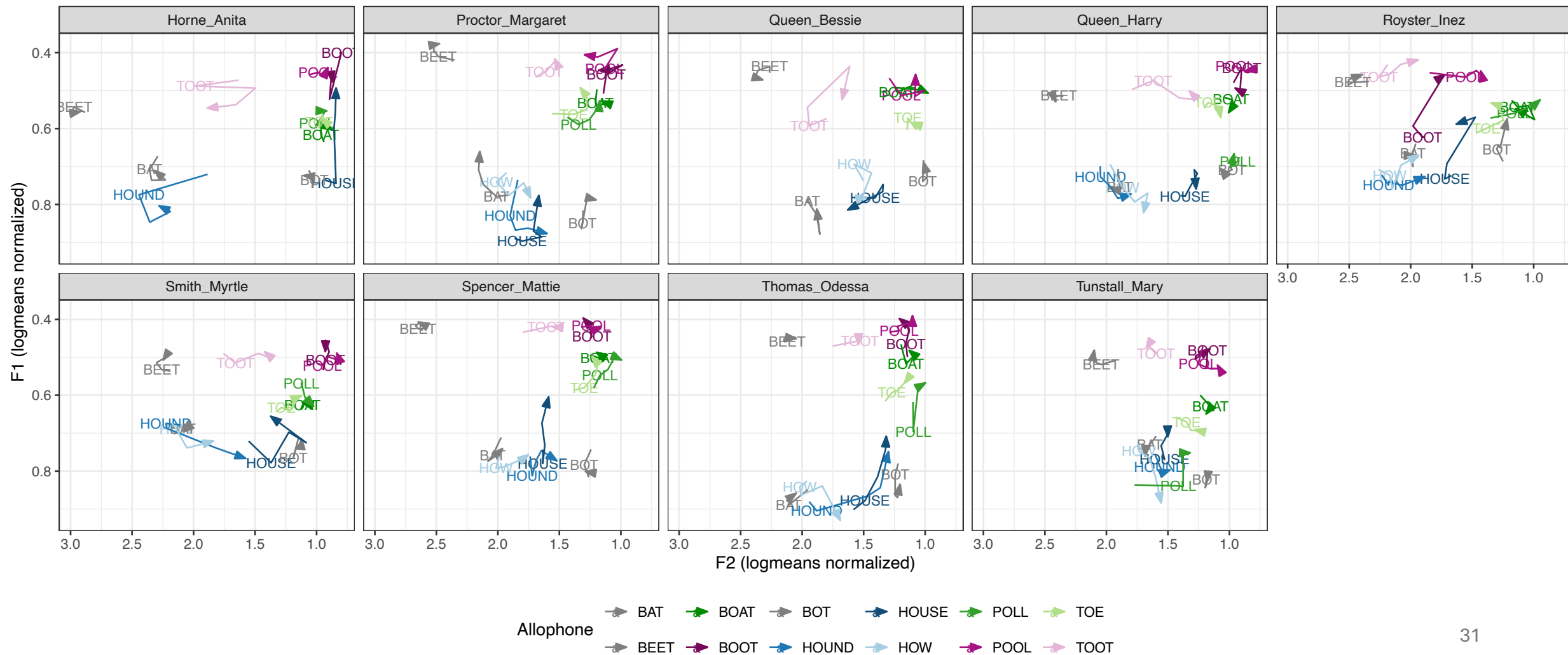


wells — GOAT ... GOOSE -- MOUTH

East Baltimore speakers: trajectories

Median trajectories by speaker and by allophone

East Baltimore speakers

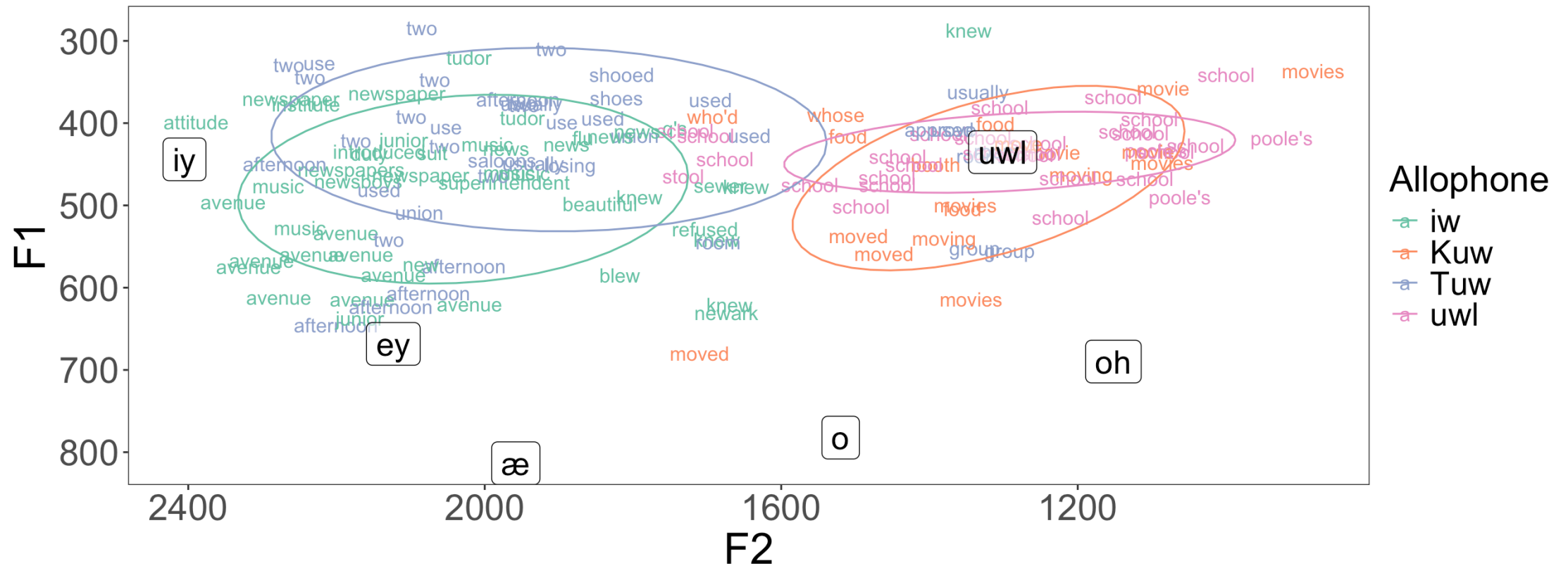


East Baltimore: more details (interviews 1979)

Name	gender	yob	age	education	ethnicity	birthplace	religion
Mary Tunstall	F	1893	85	11-12 years	Black	Baltimore County	Baptist
Bessie Queen	F	1900	78	0-7 years	Black	Baltimore	Catholic
Mattie Spencer	F	1900	78	13-16+ years	Black	VA	Christian
Margaret Proctor	F	1904	74	0-7 years	Black	NC	Baptist
Inez Royster	F	1905	73	11-12 years	Black (?)	VA	Baptist
Myrtle Smith	F	1907	71	11-12 years	Black	Baltimore	Methodist
Odessa Thomas	F	1911	67	11-12 years	Black	Baltimore	Baptist
Anita Horne	F	1918	60	11-12 years	Black	Baltimore	Methodist
Harry Queen	M	1901	77	0-7 years	Black	Baltimore	Catholic

Distinct /iw/?

Elmira Deitrich: GOOSE



And with that attitude, you don't try to teach 'em.



A pint of milk for two cents.